LoGST Architecture — Coordinate-Invariant Local-Global Spatial Transformer

Task: Predict gene expression at every tissue spot from H&E patch features.

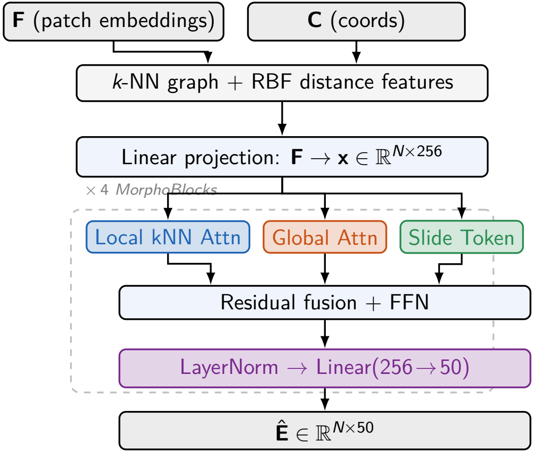
Input (per slide):

- $\mathbf{F} \in \mathbb{R}^{N \times 1536}$ — frozen UNI+CONCH patch embeddings
- $\mathbf{C} \in \mathbb{R}^{N \times 2}$ — spot pixel coordinates

Output: $\hat{\mathbf{E}} \in \mathbb{R}^{N \times 50}$ — predicted log-normalised expression

Coordinate invariance:
Coordinates enter *only* via pairwise distances (RBF-encoded).
$\Rightarrow$ invariant to translation, rotation & reflection.

Single forward pass — 4.88 M params



Results — Intra-Cohort (Table 1) & Cross-Organ LOOO (Table 2)

Table 1: Intra-cohort PCC per organ (HEST-1k)

All methods: frozen UNI+CONCH, 50 genes, 3 seeds

Cohort	ST-Net	Hist2ST	HyperST	BLEEP	STEM	LoGST
IDC (Breast)	0.551	0.309	0.567	0.519	0.439	**0.581**
COAD (Colorectal)	0.242	0.295	0.285	0.314	0.177	**0.318**
CCRCC (Kidney)	0.163	0.184	**0.256**	0.175	0.123	0.252
LUNG	0.518	0.470	0.587	0.562	0.423	**0.589**
PAAD (Pancreas)	0.359	0.247	0.404	0.419	0.233	**0.437**
PRAD (Prostate)	0.294	0.333	0.384	0.348	0.207	**0.410**
SKCM (Skin)	0.665	0.522	0.653	0.645	0.491	**0.707**
Average	0.399	0.337	0.448	0.426	0.299	**0.470**

+6.6% relative PCC gain over best baseline (HyperST)

Table 2: Cross-organ LOOO PCC per organ

Trained on 6 organs, tested on held-out organ

Cohort	ST-Net	Hist2ST	HyperST	BLEEP	LoGST
IDC (Breast)	0.198	0.458	0.465	**0.529**	0.520
COAD (Colorectal)	0.420	0.670	0.669	0.662	**0.729**
CCRCC (Kidney)	0.068	0.055	0.106	0.095	**0.109**
LUNG	0.493	0.378	0.657	0.624	**0.675**
PAAD (Pancreas)	0.459	0.399	0.463	0.537	**0.548**
PRAD (Prostate)	0.079	0.098	0.114	0.118	**0.118**
SKCM (Skin)	0.520	0.736	0.725	**0.748**	0.728
Average	0.319	0.399	0.457	0.473	**0.490**

LoGST leads on 5/7 organs under organ-shift transfer

Results — Ablation (Table 7) & Coordinate Invariance (Table 8)

Table 7: Factorial Context Ablation (Intra-cohort)

L=Local attention, G=Global attention, S=Slide token

Config	L	G	S	SKCM	LUNG	Avg
C000 (MLP only)				0.639	0.577	0.608
C001			•	0.633	0.578	0.605
C010		•		0.641	0.577	0.609
C011		•	•	0.639	0.583	0.611
C100	•			0.693	0.587	0.640
C101	•		•	0.695	0.585	0.640
C110	•	•		0.693	0.584	0.638
C111 (LoGST)	•	•	•	**0.696**	**0.589**	**0.643**

Local attention (+0.032 avg) is the dominant single contributor.

All three components together achieve the best performance.

Table 8: Numerical Coordinate-Invariance Check

Randomly initialised model, 257-spot synthetic slide.

Max. absolute output difference after coordinate transformation.

Transformation	Max. $\|\Delta\|$
Translation	5.97×10^{-5}
Rotation	1.97×10^{-5}
Reflection	0
Spot permutation	5.96×10^{-7}

Differences are numerical noise ($< 10^{-4}$), confirming predictions are invariant to how the tissue slide is oriented or positioned.

Why it matters: Two labs imaging the same tissue with different coordinate origins or rotated